PGR111, solutions week 09

Task 1 – Normalization

a) & b) together:

For the rules, see the corresponding slides.

(Below underscore means PK, asterisk means FK. This is the normal way to show normalization.)

Normalized to 3NF:

Grade (coursecode*, studentnumber*, examdate*, examresult)

Course (coursecode, coursetitle)

Student (studentnumber, studentname)

Exam (coursecode*, examdate, examregistrations)

(*Note:* the excel-sheet below shows *tables + data*, this is *not* the normal way to show normalization!)

Week 09, task 1 – Normalization

Grade

PK	PK	PK	
FK			
	FK		
FK		FK	
coursecode	studentnumber	examdate	examresult
DB1100	11111	2010-10-10	B
DB1100	11111	2012-11-11	A
DB1100	22222	2010-10-10	C
DB1100	22222	2012-11-11	A
DB1100	33333	2010-10-10	A
TK1100	11111	2010-09-09	D
TK1100	33333	2010-09-09	D
TK1100	44444	2010-09-09	E

a) Write down the rule for the second normal form (2NF). Then normalize the table to 2NF:
Show which tables you end up with, and which primary and foreign keys the tables contain.
Describe any prerequisites for your result.
(See slides or textbook for the 2NF rule.)
Prerequisites: 1) I assume that several exams can be held on the same date.
2) I also assume that a student can take several exams on the same date.
The figures in this excel sheet shows the table normalized to 2NF.

b) Write down the rule for the third normal form (3NF). Then normalize the table to 3NF:
Show which tables you end up with, and which primary and foreign keys the tables contain.
Describe any prerequisites for your result.
(See slides or textbook for the 3NF rule.)
All the 2NF tables are already on 3NF as well.

Course

PK	
coursecode	coursetitle
DB1100	Databaser
TK1100	Digital Teknologi

Student

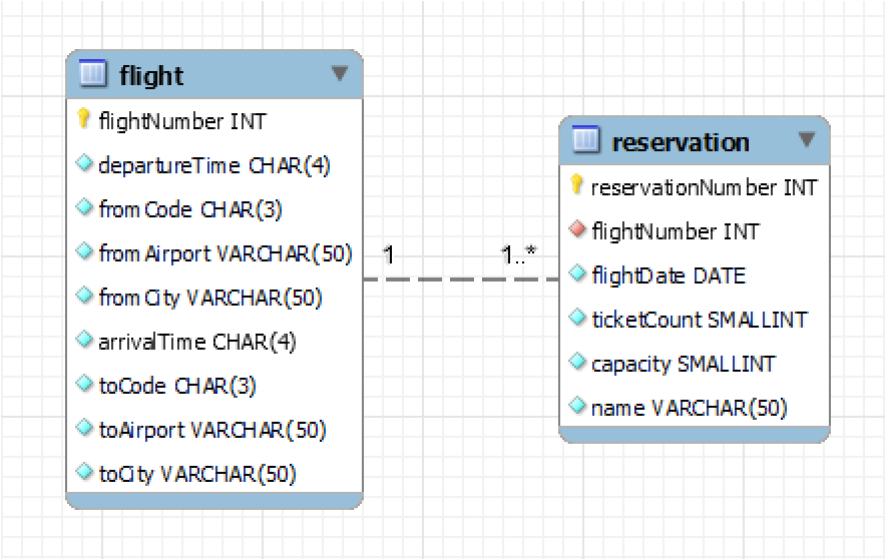
PK	
studentnumber	studentname
11111	Ola Olsen
22222	Jens Jenssen
33333	Ida Olsen
44444	Ola Olsen

Exam

PK	PK	
FK		
coursecode	examdate	examregistrations
DB1100	2010-10-10	178
DB1100	2012-11-11	17
TK1100	2010-09-09	197

Task 2 – ER modelling

ER Model for the version in the script:



Task 3 – SQL

a) Run the script as explained in the task description and in previous lectures. :-)

b)
```
INSERT INTO flight VALUES
(1, '0930', 'FBU', 'Fornebu', 'Oslo', '1040','HTR', 'Heathrow', 'London'),
(2, '1835', 'HTR', 'Heathrow', 'London', '2050','FBU', 'Fornebu', 'Oslo');

INSERT INTO reservation VALUES
(NULL, 1,'2022-12-30', 2, 130, 'Pedersen'),
(NULL, 1,'2022-12-30', 1, 130, 'Olsen'),
(NULL, 1,'2022-12-30', 4, 130, 'Andreasen'),
(NULL, 1,'2022-12-30', 1, 130, 'Karlsen'),
(NULL, 2,'2022-12-31', 2, 95, 'Pedersen'),
(NULL, 2,'2022-12-31', 1, 95, 'Jonasen'),
(NULL, 2,'2022-12-31', 1, 95, 'Fransen'),
(NULL, 2,'2022-12-31', 4, 95, 'Andreasen');
```

c)
```
UPDATE flight
SET fromCode = 'OSL', fromAirport = 'Gardermoen'
WHERE fromCode = 'FBU';

UPDATE flight
SET toCode = 'OSL', toAirport = 'Gardermoen'
WHERE toCode = 'FBU';
```

Task 4 – Normalization

a) See the slides on unnormalized tables/DBs from todays lecture.

b) Determinants:

-- table flight --

flightNumber → departureTime, fromCode, fromAirport, fromCity, arrivalTime, toCode, toAirport, toCity
fromCode → fromAirport, fromCity
toCode → toAirport, toCity

-- table reservation --

reservationNumber → flightNumber, flightDate, ticketCount, capacity, name
flightNumber, flightDate → capacity

NOTE: In table flight, fromCode & toCode contain the same type of data, and are determinants for the same kind of data (fromAirport & toAirport, fromCity & toCity). By combining these (drop "from" and "to" prefixes), we omit what would otherwise be 2 tables with identical data. Instead we make 2 relationships between the same tables (one for departure, one for arrival).

Normalized to 3NF, we get the following tables: (underscore = PK)
(*NOTE:* We also use this opportunity to give some of the columns more fitting names, and not being a fan of PKs consisting of several columns, we give flightPlan a surrogate PK)
flight (number, departureTime, arrivalTime, fromAirport_code*, toAirport_code*)
airport (code, name, city)
reservation (number, customerName, ticketCount, flightPlan_id*)
flightPlan (id, date, capacity, flight_number*)

c) ER model for the 3NF version:

